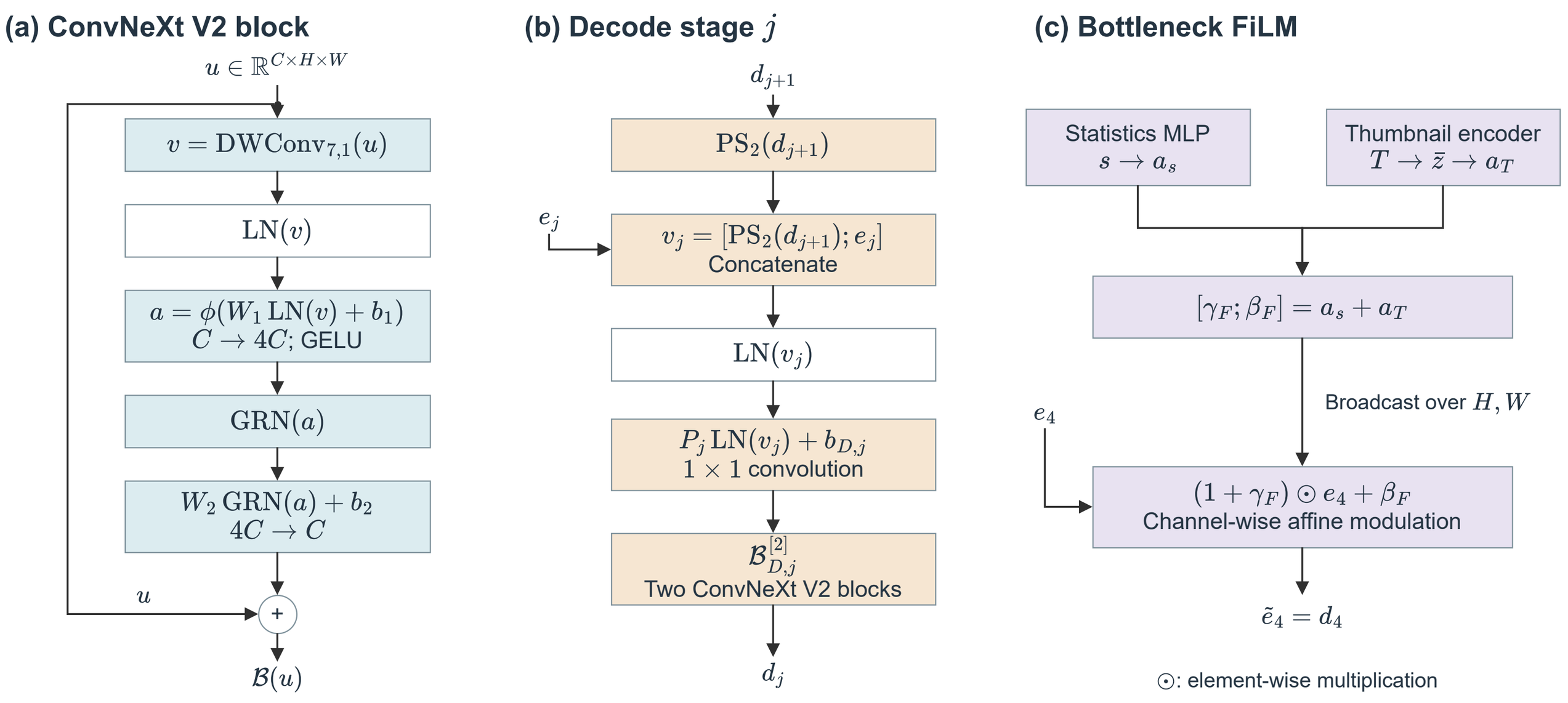

(a) ConvNeXt V2 block
$u \in \mathbb{R}^{C\times H\times W}$
$v = \mathrm{DWConv}_{7,1}(u)$
$\mathrm{LN}(v)$
$a = \phi(W_1\,\mathrm{LN}(v) + b_1)$
$C \to 4C$; GELU
$\mathrm{GRN}(a)$
$W_2\,\mathrm{GRN}(a) + b_2$
$4C \to C$
u
+
$\mathcal{B}(u)$
(b) Decode stage j
d_{j+1}
$\mathrm{PS}_2(d_{j+1})$
e_j
$v_j = [\mathrm{PS}_2(d_{j+1}); e_j]$
Concatenate
$\mathrm{LN}(v_j)$
$P_j\,\mathrm{LN}(v_j) + b_{D,j}$
1×1 convolution
$\mathcal{B}^{[2]}_{D,j}$
Two ConvNeXt V2 blocks
d_j
(c) Bottleneck FiLM
Statistics MLP
$s \to a_s$
Thumbnail encoder
$T \to \bar{z} \to a_T$
$[\gamma_F; \beta_F] = a_s + a_T$
Broadcast over H, W
e_4
$(1 + \gamma_F) \odot e_4 + \beta_F$
Channel-wise affine modulation
$\tilde{e}_4 = d_4$
$\odot$: element-wise multiplication